On Market Making for a New Exchange: EUR/USD Designated Market Maker on Exchange A

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1 Introduction

1.1 Motivation

A newly launched exchange — referred to throughout this report as exchange A — has mandated our firm as the sole Designated Market Maker for its EUR/USD spot book. Two established venues, B and C, remain the primary sites of price discovery, carrying on average 75% and 25% of the daily volume respectively. We receive market data from B with a 200 ms delay and from C with a 170 ms delay, while high-frequency participants reach the same data in 50 ms and can act on exchange A before our own view of fair value has refreshed. The mandate commits us to continuously quoting ten levels on each side of the book at a common 1 bp tick, with a starting capital K split evenly between EUR and USD and an asymmetric fee grid. A delta limit caps the EUR inventory we are willing to warehouse before externalising risk as taker flow on B and C.

Three constraints shape every design decision that follows. First, on the day exchange A goes live no historical fills or counterparty behaviour are observable, so the initial quoting policy cannot be learnt from data and must be derived from first-principles microstructure models. Second, the latency gap to HFTs exposes every resting quote to adverse selection, which the pricing model must compensate for rather than ignore. Third, because we cannot market-make on B or C, the delta limit turns inventory management into an active cross-venue hedging problem rather than a passive book-balancing exercise.

1.2 Objective

Our goal is to deliver an end-to-end market-making system that addresses the three constraints above and can be evaluated under the conditions described in the subject brief. The project is split into three phases matching the mandate timeline. Phase 1 bootstraps liquidity without any trading history, using a heuristic quoting policy derived from classical market-making theory. The fair mid is reconstructed from the two reference venues at their respective latencies, the bid-ask spread is sized to compensate for both inventory risk and adverse selection, and quotes are skewed in the direction of recent order-flow imbalance. A dedicated hedge router externalises excess EUR inventory on B and C whenever the delta limit is approached. Phase 2 re-estimates the Phase 1 parameters by maximum likelihood from a simulated month of fills. Phase 3 studies how our quoting policy should degrade from a price-competitive quoter to a reliable fallback liquidity provider once aggressive HFT market makers populate the inside of the book.

1.3 Simulation Environment

Because the exchange has no history at launch and we cannot run the strategy against real counterparties, the evaluation environment is a layered simulation built specifically for this project. A stochastic mid-price generator (GBM, GARCH with jumps, or Heston) produces the fair value underlying both reference venues. A venue simulator wraps that path into a realistic bid/ask and depth profile per venue, with a choice of five spread regimes (static, stochastic, adaptive, asymmetric, and skewed) reflecting the microstructure assumptions we want to stress. A limit order book with a matching engine and fill events sits at the centre of exchange A; a Poisson client-flow generator with truncated Pareto sizes delivers counterparty orders whose intensity decays exponentially in

the quoted half-spread; and a reporting layer reconstructs the full mark-to-market and realised P&L decomposition from the trade history.

1.4 Evaluation

Each phase is evaluated on the subject-mandated one-month simulated backtest, with the capital K reset to its initial split at the start of every phase. For Phase 1 we report realised and mark-to-market P&L, the percentile envelope of mark-to-market as a function of time since trade inception, top-ten trades by size, full-versus-partial fill rates at each of the ten quoting levels, and inventory trajectories against the delta limit. For Phase 2 we compare the heuristic and calibrated configurations on an independent month using the same diagnostics, so that any difference in performance is attributable to the updated parameters rather than to the market regime. For Phase 3 we measure both our own P&L under HFT competition and the quality of liquidity observed by exchange-A clients, distinguishing small from large trades and quantifying the continuity of two-sided quoting through volatility spikes, one-sided HFT retreats, and venue outages.

1.5 Report Structure

The remainder of the report follows the step-by-step structure of the subject. section 2 introduces the order-book data structure and matching rule; section 3 describes the venue simulator and the family of spread regimes supported; section 4 presents the quoting framework in full; section 5 covers submission and cancellation handling; section 6 documents the simulation loop that binds the modules together; section 7 details the inventory and P&L bookkeeping; section 8 gathers the Phase 1 backtest outputs; section 9 analyses the calibration results; section 10 examines the Phase 3 fallback behaviour; and section 11 summarises the findings and outlines the remaining modelling questions.

2 Order Book

The order book of exchange A must show a price and a quantity at ten levels on both sides at all times. This section describes the data structure we chose to hold that state, how submissions and cancellations flow through it, and the invariants of the matching engine. The design is exercised on a realistic EUR/USD seed (mid = 1.10, 1 bp tick, ten levels per side, 1M EUR per level).

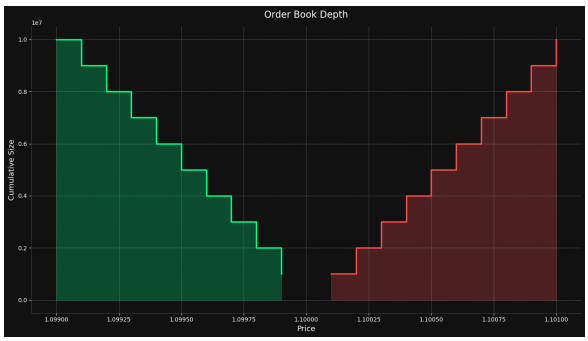
2.1 Data structure

Two pressures shape the design. Matching requires a price-time-ordered view of the market-maker (MM) side at clearing time, yet the hot path is dominated by bulk requotes — up to ten cancels followed by ten posts per step. Keeping the book permanently sorted would pay on every mutation for a benefit consumed only once per step.

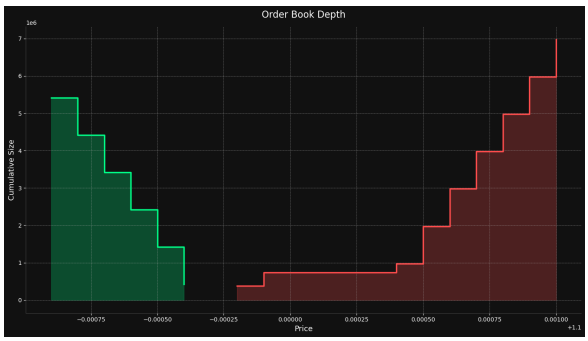
We therefore split orders by origin (MM versus client) into two hash-indexed stores and keep a price-ordered view of the MM side as a lazy cache, rebuilt only when the matcher runs. Cancels and posts touch only the hash stores; the MM side is sorted once per clearing call instead of being re-ordered on every mutation. Time priority within a price level is preserved by a counter that grows by one on every insertion, so later orders carry higher numbers than earlier ones.

2.2 Submissions and cancellations

The two sides are handled asymmetrically. MM cancels and posts arrive in bulk and never run the matcher, because passive quotes cannot cross by construction. Client orders are inserted one at a time, and the matcher runs immediately after each insertion.



(a) Seeded book: one order per level, symmetric depth.



(b) After twenty marketable client orders: shallow levels consumed from the top.

Figure 1: Cumulative depth before and after a burst of client flow.

2.3 Matching rule

Three invariants govern every clearing call. Client orders only match against MM orders, never against each other: as the sole designated market maker for Phases 1 and 2 we are on one side of every trade. The fill price is always the passive MM price, never the incoming aggressive price — clients pay the spread we quote, which is the only source of gross revenue. And matching follows strict price-time priority with partial fills: the engine walks the sorted MM side, deducts the minimum of the two sizes, logs the match,

and either retires the filled order or leaves the remnant resting. The residual shape in Figure 1 is a direct visual check of this rule: the top of the book is consumed first, deeper levels are untouched.

2.4 Fills as events

Every match is dispatched synchronously, as a fill event, to a single registered listener. In the full simulation the listener is the Quoter, which updates inventory, trade history, and the pending top-up queue inside the clearing call. The Quoter therefore never polls the book: the next quote computation already sees the post-fill inventory, which removes staleness bugs on steps that both fill and requote. Decoupling the book from its consumer also keeps the same engine usable in Phase 3, where a separate diagnostic listener records fills for the exchange-client-side liquidity analysis without touching the Phase 1 code path.

3 Market Simulator for Venues B and C

Exchanges B and C are external reference venues: we never post on them, we only read them. Our quoting engine on A consumes the best bid, the best ask, and the best-size of each, and nothing else. We therefore do not simulate a full limit order book on B or C: all the depth beyond the top of book would be detail the rest of the system never looks at. Instead we simulate a mid-price process and build a synthetic bid/ask/depth layer around it. This is the second of the two options offered by the subject, and it is both cheaper to run and sufficient for everything the market maker on A has to decide. The reduced-form choice keeps realism where it matters — spreads that widen under stress and depth that thins with them — while avoiding the combinatorial cost of simulating individual order arrivals at two venues.

3.1 Noised mid

The fundamental value V_t is the GBM (or GARCH, or Heston) path produced by the stock simulator. The observed mid on B and C is V_t plus microstructure noise, in the spirit of Roll [7]:

$$P^{\text{mid}}_{B/C}(t) = V_t + \eta \cdot \sigma \cdot \sqrt{\Delta t} \cdot Z_t, \quad Z_t \sim \mathcal{N}(0,1),$$

with $\eta = 0.1$. Each venue draws an independent Z_t path, so B and C are noised versions of the same fundamental. One subtle point: every spread model below estimates realized volatility from V_t, not from P^{mid}, because differencing the noised series would double-count the microstructure variance inside the vol estimator. The zero-mean estimator used for that (see `compute_rv_zero_mean`) is unbiased for zero-drift GBM and lower-variance than the sample std over short windows.

3.2 Spread regimes

We benchmarked five regimes of increasing realism — Static, Stochastic, Adaptive, Asymmetric, and Skewed — and kept two. We reject Static because a price-independent width ignores volatility entirely; the market maker is widest exactly when they should be widest, but never adapts. Stochastic scales the width with realized vol but never returns to the competitive floor even in calm markets, which systematically over-charges. Asymmetric mean-reverts realistically but treats up-moves and down-moves the same way, leaving a latency-arbitrage gap for HFTs who, seeing the vol spike 150 ms before us, can infer our quote path. We deploy **Adaptive on** C and **Skewed on** B.

Venue C — **Adaptive.** A symmetric half-spread proportional to how much realized vol deviates from the long-run input σ:

$$h(t) \;=\; h_0 \cdot \Big(1 + \alpha \cdot \big(\widehat{RV}_t/\sigma - 1\big)\Big), \qquad h(t) \geq 0.1 \cdot h_0,$$

with $h_0 = \text{mid}(t) \cdot s_{\text{bps}}/10000$ and $\widehat{RV}_t$ an EMA-smoothed rolling vol over a 120 s window. The width is tight at baseline, widens on shocks, and tightens symmetrically when calm returns. A 10% floor prevents collapse near zero. C carries 25% of volume — a smaller venue where informed flow is weaker, competition enforces a tight floor, and the symmetry of up vs. down adjustment is not a material leak.

Venue B — Skewed. Same vol-adaptive width, plus a directional shift of the quote centre driven by a short-horizon momentum signal. Let $\hat{m}_t$ be the EMA of log-returns normalised by its steady-state noise floor; the skew is

$$\delta_t \;=\; \gamma \cdot h(t) \cdot \tanh\big(\mathrm{sign}(\hat{m}_t) \cdot \max(|\hat{m}_t| - \tau,\, 0)\big),$$

with $\tau = 1.5$ the dead-band threshold. Small random wiggles map to $\delta_t = 0$ exactly; only sustained directional moves shift the centre. Posted quotes become ask $=$ mid $+ h + \delta$ and bid $=$ mid $- h + \delta$: width is unchanged, the centre moves in the direction of the trend. B carries the dominant 75% of volume, where short-horizon order-flow imbalance predicts returns [3]; a momentum-unaware venue there would be repeatedly picked off [2, 4].

3.3 Depth

Depth at each step is derived deterministically from spread width, not simulated as an independent process:

$$\mathrm{depth}_t \;=\; \bar{D} \cdot \frac{\overline{S}}{S_t},$$

where S_t is the full spread and $\overline{S}$ its session mean. The economic story is simple: when the external market maker is confident, they quote tight and large; when stressed, they pull size and widen. The hedge router on A reads this array with the same 200 ms (B) / 170 ms (C) latency offset as the mid, and caps each hedge leg at the available depth.

3.4 One stock, two venues

B and C are two `Market` objects wrapping the same `Stock` via `copy.deepcopy`. The fundamental path V_t is shared; the noise draws and spread realisations are independent. We deliberately do not model B/C price divergence: the cross-venue edge in this project is the latency asymmetry (HFTs see B/C at 50 ms while we see them at 200/170 ms), not price disagreement between venues.

3.5 Validation

Figure 3.5 shows the five width paths under a GARCH-plus-jump run — the one regime where the models visibly differentiate. Each width panel is annotated with its Pearson correlation to realized vol; the bottom panel shows the skew δ_t activating at each jump under the Skewed regime and remaining at zero elsewhere.

[Figure placeholder]
spread-regimes-garch.png — export from
`market_garch.compare_spreads()` in `test/price_simulation.ipynb`

Figure 2: Width panels with realized-vol overlay and the Skew δ_t path, GARCH+jump run.

4 Market-Making Logic on Exchange A

Exchange A has no history, so every quote we post is derived from the state of B, C, our inventory q, and our capital K. Given these inputs the quoter must output two things each step: the set of orders to post and the set of resting orders to cancel. We describe here the modelling decisions that produce those two lists, not the Python that assembles them. The six subsections follow the order in which the decisions are composed: read the reference price, compute a reservation price, widen a spread around it, skew asymmetrically and lay down a ladder, decide whether to re-quote at all, and finally hedge when inventory breaches its limit.

4.1 Fair mid under latency asymmetry

We never quote against A's own mid: A has no informative price until we seed one. The reference is instead a cross-venue best constructed from B and C read with their respective latency offsets $\ell_B = 200\,\text{ms}$ and $\ell_C = 170\,\text{ms}$,

$$\text{bid}_t^{\text{fair}} = \max\big(\text{bid}_B(t-\ell_B),\ \text{bid}_C(t-\ell_C)\big), \qquad \text{ask}_t^{\text{fair}} = \min\big(\text{ask}_B(t-\ell_B),\ \text{ask}_C(t-\ell_C)\big),$$

with $\text{mid}_t^{\text{fair}} = \frac{1}{2}(\text{bid}_t^{\text{fair}} + \text{ask}_t^{\text{fair}})$. The choice of the cross-venue *best* rather than a volume-weighted average is a defence against being arbitraged: clients can always transact at the tighter side elsewhere, so anything wider on A is free optionality handed to adversarial flow. HFTs read the same feeds at $\ell_{\text{HFT}} = 50\,\text{ms}$, so the price we quote against is always structurally stale relative to theirs; this gap is priced explicitly as a spread component in Section 4.3.

4.2 Reservation price: Avellaneda–Stoikov with infinite horizon

The classical inventory-shift mechanism of Avellaneda and Stoikov [1] uses a finite horizon T with the penalty $\gamma\sigma^2(T-t)$. That formulation is inappropriate for a 24/5 FX market: there is no terminal liquidation time at which inventory becomes worthless. We replace $T-t$ by a stationary discount $1/\omega$ with $\omega = 1/(8\,\text{h})$ — one FX session as the effective risk horizon — which is the infinite-horizon limit of the same HJB system, studied formally by Guéant et al. [6].

A second correction is numerical. Applying the raw Avellaneda–Stoikov shift with inventory in EUR (range 0–900k) produces a reservation displacement of several hundred basis points at ten percent inventory, three orders of magnitude larger than the half-spread itself. We therefore work with the normalised ratio $\rho = q/K$ and express the shift as a fraction of the half-spread, so that at $|\rho| = 1$ the quoted centre is displaced by exactly one half-spread:

$$r_t \;=\; \text{mid}_t^{\text{fair}} \;-\; \rho \cdot h_t.$$

Direction is preserved (a long EUR position drags the centre down and attracts sells); magnitude is now calibrated to the spread itself, with no new free parameter beyond γ.

4.3 Three-component spread with fee floor

The half-spread h_t is built as the sum of three economically distinct adverse-selection premia, floored by a fee-cover term,

$$h_t \;=\; \tfrac{1}{2}\big(h_t^{\text{AS}} + h_t^{\text{lat}} + h_t^{\text{inv}}\big), \qquad h_t \;\geq\; \phi \cdot \text{fee}_{\text{maker}}^{A} \cdot \text{mid}_t^{\text{fair}}.$$

The Avellaneda–Stoikov component

$$h_t^{\mathrm{AS}}(\mathrm{bps}) \;=\; \gamma \cdot (\sigma \cdot 100)^2 \cdot \tfrac{1}{\omega} \;+\; \tfrac{2}{\gamma}\,\ln\!\big(1+\gamma/k\big)$$

combines inventory-risk aversion (first term) with the convexity of the exponential order-arrival intensity (second term), as derived by Avellaneda and Stoikov [1] after the infinite-horizon substitution. The latency term $h_t^{\mathrm{lat}} = 2\sigma_{\mathrm{sec}}\sqrt{\Delta\ell}$, with $\Delta\ell = w_B(\ell_B - \ell_{\mathrm{HFT}}) + w_C(\ell_C - \ell_{\mathrm{HFT}})$, is two standard deviations of the expected adverse move during the window in which HFTs see the reference feeds and we do not — a cushion against the free option they hold against our stale quotes. The capacity term $h_t^{\mathrm{inv}} = \alpha_s \rho^2 h_t^{\mathrm{AS}}$ is quadratic in the inventory ratio: it is negligible near flat and widens sharply as we approach the delta limit, pushing flow away from the side that would deepen our position. Finally the floor $\phi \cdot \mathrm{fee}^A_{\mathrm{maker}} \cdot \mathrm{mid}_t^{\mathrm{fair}}$ guarantees that the inception spread covers the maker fee on every fill: no trade is ever gross-negative, only fee-negative offset by subsequent inventory P&L. The floor binds in calm regimes where the three adversarial premia would otherwise collapse below the fee.

4.4 Asymmetric skew and ladder

The reservation shift in Section 4.2 is symmetric: it moves both sides of the quote by the same amount. Guéant [5] shows that the asymptotic optimum is instead an *asymmetric* split of the half-spread — widen the side that would deepen inventory, tighten the side that would reduce it. We therefore apply the Guéant skew

$$\Delta_t \;=\; q \cdot \sqrt{\frac{\gamma\sigma^2}{2k}}, \qquad \mathrm{ask}_t^{\mathrm{post}} = r_t + h_t - \Delta_t, \qquad \mathrm{bid}_t^{\mathrm{post}} = r_t - h_t - \Delta_t,$$

with Δ_t clipped so the tighter side can never breach the fee floor.

The subject requires ten levels on each side. Levels step one tick further from the mid at each rung; sizes decay exponentially and are skewed against inventory,

$$\mathrm{size}_i^{\mathrm{bid}} \;=\; Q_0\, e^{-\beta i}\,(1 - \tfrac{1}{2}\overline{\rho}), \qquad \mathrm{size}_i^{\mathrm{ask}} \;=\; Q_0\, e^{-\beta i}\,(1 + \tfrac{1}{2}\overline{\rho}),$$

with $\overline{\rho} = \mathrm{clip}(\rho, -1, 1)$. This is a second, complementary inventory correction acting through *depth* rather than *price*: even if the Guéant skew is clipped by the fee floor, the ladder sizes still bias against further accumulation. Exponential decay concentrates displayed liquidity at the top of book, consistent with the empirical shape of limit-order-book queues.

4.5 Session-adaptive arrivals and selective requoting

EUR/USD trades across three active sessions — Tokyo, London, New York — with markedly different depth profiles. A single arrival decay k calibrated on one session misprices fill probability on the others, since the same quoted distance attracts very different order flow under different liquidity regimes. We therefore apply session multipliers to k (London deepest, Tokyo thinnest, New York as reference) and treat the 08:00 and 16:00 UTC transitions as hard resets of the full quote ladder so session-specific state does not bleed across regimes.

Re-quoting every step is expensive in cancel/post messages and costly in time priority: orders that are still valid lose their queue position on every spurious cancel. We use a strict

six-priority hierarchy, with each lower priority firing only if none above it did (Table 1). Priority 3 closes the price-drift leak; priority 5 closes a subtler silent-accumulation leak where market prices are flat but our own inventory has moved enough to invalidate the ladder skew. Priority 6 fires on pure staleness and re-quotes at a tighter 0.7× spread, so the refreshed quotes compete back into relevance rather than being re-posted at the same passive distance.

P	Trigger	Action
1	First call	Full ladder, no cancels
2	Session boundary (08:00/16:00 UTC)	Cancel all, full ladder
3	$\lvert$best-price drift$\rvert > 0.25$ spread	Cancel all, full ladder
4	Fills / top-ups pending	Rebuild filled slots only
5	$\lvert\Delta q\rvert/K > 5\%$ since last reprice	Cancel all, full ladder
6	Order older than τ_{stale}	Cancel stale, reprice at 0.7× spread

Table 1: Requote priorities; at most one rule fires per step.

4.6 Hedge router

When either $|q| > \delta_{\text{lim}} \cdot K/2$ or $|q \cdot \text{mid}^{\text{fair}}| > \delta_{\text{lim}} \cdot K/2$ — that is, either the EUR leg or its USD notional breaches the delta limit on half-capital — we externalise risk through taker market orders on B and C. Three decisions shape the router. First, the USD notional of EUR inventory is valued at the B/C cross-venue mid, not at mid_A, because A's mid is polluted by our own skewed quotes. Second, the split between venues is depth-aware: each venue is scored by

$$s_v \;=\; \frac{1}{\text{fee}^v_{\text{taker}} \;+\; \sigma_{\text{sec}} \cdot \ell_v \;+\; \lambda \cdot \text{mid}^{\text{fair}}/D_v},$$

combining the taker fee, the expected adverse move during execution lag, and a market-impact penalty inversely proportional to the available depth D_v at the lagged step. The hedge is split $s_B : (s_B + s_C)$ with overflow re-routed to the other venue when the preferred one is depth-constrained. Third, when post-hedge inventory remains above the tighter threshold hedge_partial_limit$\cdot K/2$ — meaning the combined $B{+}C$ depth was insufficient to bring us back — we flip an emergency flag that multiplies the Avellaneda–Stoikov inventory-risk term by 5× on the next step. Our own quotes on A become so aggressively asymmetric that organic client flow cleans up the residual. This is the Guéant et al. [6] inventory-penalty mechanism used as an overflow valve: we deliberately do not chase depth we cannot reach.

5 Applying Submissions and Cancellations

6 Simulation Loop

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8 Phase 1 Backtest Report

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